Security and Privacy for Machine Learning

IA-2: Implementation, Evaluation and Technical Outcomes

# Membership Inference as a Common Yardstick: Evaluating Five Families of Privacy-Preserving Defences for Healthcare Machine Learning

Final Technical Report (individual submission)

| Field | Details |
|---|---|
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| Roll number | 16014223012 |
| Batch | B1 |
| Assessment | IA-2 (20 marks) - continuation of IA-1 research foundation |
| IA-1 topic | Membership Inference Attack on ML Models: A Healthcare Privacy Use Case |
| Datasets | Pima Indians Diabetes (primary); Breast Cancer Wisconsin Diagnostic (replication) |
| Code & data | Project folder `mlcs ia/` (src/, data/, results/) - see Appendix A |
| Date | 24 October 2026 |

*All results in this report were produced by the code in* `src/` *and are read directly from* `results/tables/`*; no value was typed in by hand. Experiments use only public datasets and locally trained models - no real person, service or organisation was attacked.*

# Abstract

Machine-learning models trained on medical records can reveal whether a particular patient's record was part of the training set. IA-1 framed this as a black-box **membership inference attack (MIA)** on a diabetes-risk classifier. IA-2 turns that attack into a **common yardstick** and uses it to evaluate one defence family from every course module: regularisation and model choice (Module 1), central differential privacy via DP gradient descent with a zCDP accountant (Module 2), local differential privacy with the Piecewise mechanism and LDP Naive Bayes (Module 3), Mondrian k-anonymity, l-diversity, input randomisation and output restriction (Module 4), and compressive privacy with PCA/DCA/DUCA projections (Module 5). Each configuration is trained as a target model plus eight shadow models over 10 random splits, attacked with a shadow-model attack and a loss-threshold attack, and compared with the overfitting baseline using paired Wilcoxon tests.

On the Pima data the baseline MLP leaks clearly (shadow-attack AUC 0.765, train-test gap 0.281). Strong regularisation removes most of the leakage (AUC 0.494) with no significant change in accuracy, and a plain logistic regression is both the most accurate and essentially leak-free model. DP-GD at $\varepsilon = 4$ holds the worst-case attack AUC at 0.511 with 74.7% test accuracy (baseline 71.9%), and the attacker's measured advantage stays well below the theoretical DP bound. Restricting the API to labels only lowers the attack AUC only to 0.666, and k-anonymity ($k = 25$) only to 0.598, so output hiding and data generalisation are not substitutes for formal guarantees. LDP gives the strongest trust model but needs far more users than this dataset has. Compressive privacy with one DCA component keeps 77.2% utility-task accuracy while cutting the attacker's balanced accuracy at inferring age from 1.00 (raw data) to 0.62. The trends replicate on the Breast Cancer dataset.


**Keywords** - membership inference, differential privacy, local differential privacy, k-anonymity, compressive privacy, privacy-utility trade-off, healthcare ML.

# 1. Problem Refinement and Objectives

## 1.1 Summary of IA-1

IA-1 (the Module 1 research-foundation document and the IA-1 report) defined a black-box membership inference attack against a diabetes-risk classifier on the Pima data. It mapped the problem to all five modules, prepared the data with public-bounds preprocessing, and ran a baseline shadow-model attack on a single split (seed 0) for models A-D. The unregularised MLP leaked clearly there (attack AUC 0.738). The DP accountant, LDP mechanisms, Mondrian partitioner and PCA/DCA projections were prototyped and checked in isolation but not yet attacked. On that split the regularised model D also collapsed towards the majority class, which showed that one split is not enough to judge a defence.

## 1.2 Revised problem statement

*How much training-set membership does a black-box attacker learn from a healthcare classifier, and which privacy-preserving techniques from the five course modules reduce that leakage, at what cost in predictive utility and under which trust assumptions?*

The refinement keeps the IA-1 attack unchanged but uses it as the **measurement instrument** for every module. This gives the five modules one shared, quantitative privacy metric instead of five unrelated demonstrations.

## 1.3 Research questions

| RQ | Question | Module |
|---|---|---|
| RQ1 | How strongly does membership leakage depend on model complexity and overfitting? | M1 |
| RQ2 | How does the central DP budget ε trade utility against measured leakage, and how does measured leakage compare with the theoretical DP bound? | M2 |
| RQ3 | Can patients protect themselves with LDP before data collection, and what data volume does that require for useful accuracy? | M3 |
| RQ4 | Do classical PPDM techniques (k-anonymity, l-diversity, input noise, output restriction) mitigate MIA, and what attacks do they leave open? | M4 |
| RQ5 | Can compressive projections keep the utility task accurate while suppressing a privacy task and reconstruction? | M5 |
| RQ6 | Across all modules, which defences give the best privacy-utility balance for this use case? | All |

## 1.4 Final objectives and technical requirements

- **O1** Implement a reproducible MIA pipeline (target + 8 shadow models, two attackers, three API modes).
- **O2** Implement at least one technique per module *from first principles* (DP-GD + accountant, LDP mechanisms + OUE, Mondrian + l-diversity, DCA/DUCA) so that each mechanism is auditable.
- **O3** Evaluate utility (accuracy, F1, ROC-AUC, generalisation gap) and privacy (attack AUC, accuracy, precision, recall, F1, advantage) for every configuration over 10 seeds.
- **O4** Test significance against the baseline with paired tests and report mean ± SD.
- **O5** Replicate on a second public medical dataset to check external validity.
- **O6** Document provenance, preprocessing, ethics and reproducibility to the IA-2 checklist.
- **Requirements:** data-independent preprocessing (needed for valid DP/LDP guarantees); Kerckhoffs-style attacker (shadow models use the defended training recipe); fixed seeds; runtime under one hour on a laptop.

## 1.5 Changes since IA-1 and their justification

| Aspect | IA-1 plan | IA-2 change | Reason |
|---|---|---|---|
| Scope | MIA + defences discussed (Module 1 focus) | One defence family per module implemented and evaluated against the same attack | IA brief requires integration of all five modules |
| Research question | Single RQ on attack effectiveness and regularisation | RQ1-RQ6 (Sec. 1.3) | Each module needs a testable question |
| Attacker | Shadow models + logistic-regression attack | Added loss-threshold attack, rounded and label-only APIs | IA-1 limitation: one simple attack may underestimate risk |
| Statistics | Single split | 10 seeds, mean ± SD, paired Wilcoxon, 95% CI | IA-1 Sec. 10 called for repeated runs |
| Data | Pima only | Pima + Breast Cancer Wisconsin replication | External-validity limitation |
| Preprocessing | Public-bounds scaling, zeros as missing, missing indicators | Kept; added public clinical bins for LDP Naive Bayes | Bins must be data-independent for the LDP guarantee to hold |
| Defences M2-M5 | Mechanisms prototyped and checked in isolation (accountant, LDP variance, Mondrian, PCA/DCA) | Each defence built into a training recipe and attacked; DP-GD in NumPy with per-example clipping; DUCA, l-diversity, input noise and output restriction added | IA-1 left the defences unattacked; own DP-GD keeps the accounting transparent (the foundation document had planned TensorFlow Privacy) |
| Model matrix | Models A-D | A-E (added deep random forest) | RF is a known high-leakage model |
| Knowledge distillation | Candidate defence | Not implemented (discussed in Sec. 4.10) | No formal guarantee; time spent on formal (DP/LDP) defences instead |

## 1.6 Scope and extended threat model

| Component | Assumption in IA-2 |
|---|---|
| Target | Binary classifier trained on 25% of the records (members); another 25% are non-members |
| Attacker knowledge | 50% of the records from the same distribution (shadow pool), the training recipe and any published transform (Kerckhoffs' principle) |
| Access | Black-box queries: full probabilities, probabilities rounded to 0.1, or label only |
| Goal (MIA) | Decide whether a known record was in the target training set |
| Goal (M4) | Link a released k-anonymous record to a person through quasi-identifiers and learn the diagnosis (homogeneity attack) |
| Goal (M5) | From compressed vectors z, infer a sensitive attribute (age >= 35) or reconstruct x |
| Trust (M3) | Data collector untrusted: each patient perturbs their own record |
| Safety | Public data, locally trained models; no real system queried, no re-identification |

## 2. Integration of Course Concepts

Figure 1 shows the end-to-end pipeline. Every defence is expressed as a *training recipe* `recipe(X, y, rng) -> model`. The same function trains the target and all eight shadow models, so the attacker always faces the defended system and no defence gets credit from a mismatched attacker.

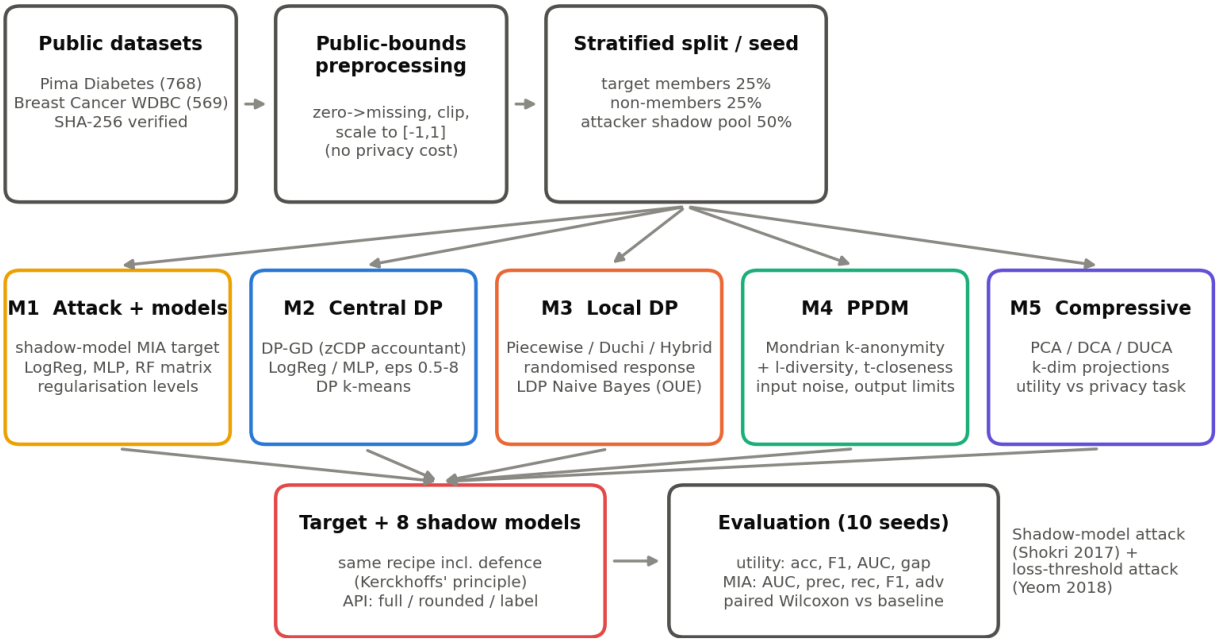



*Figure 1. IA-2 architecture: shared data pipeline, one defence family per module, a common attack harness and paired evaluation.*

### 2.1 Final module-mapping table

| Syllabus topic | How it is used in this work | Treatment | Evidence |
|---|---|---|---|
| **1.1** Privacy complications in the AI era | Healthcare use case: membership itself reveals diagnosis/treatment participation | Analytical | Sec. 1, IA-1 |
| **1.2** Threats & attacks on ML | Shadow-model MIA (Shokri) + loss-threshold MIA (Yeom); label-only and rounded API variants | Implemented | `attacks.py`, Tab. 3, Fig. 2 |
| **1.3** Privacy-preserving ML | Model-complexity / regularisation matrix A-E as the first line of defence | Experimental | `models.py`, Tab. 3 |
| **1.4** PPDM overview | Bridges to Module 4; positions the defence families by where they act (input, training, output) | Analytical | Sec. 2.2 |
| **2.1** DP mechanism & properties | Gaussian mechanism on clipped gradients; sequential composition (zCDP), post-processing, DP-to-advantage bound | Implemented + demo | `m2_dp.py`, Fig. 3-4 |
| **2.2** Applying DP in ML | DP gradient descent with per-example clipping | Implemented | Tab. 4 |
| **2.3** DP supervised learning | DP logistic regression and DP MLP-128, $\varepsilon \in \{0.5,\ldots,8\}$ | Experimental | Tab. 4, Fig. 3 |
| **2.4** DP unsupervised learning | DP k-means (DPLloyd) on Pima features | Controlled demo | Fig. 4 |
| **3.1** LDP & mechanisms | Laplace, Duchi, randomised response; variance checked against theory | Implemented + correctness check | `m3_ldp.py`, Fig. 5 |
| **3.2** Advanced LDP | Hybrid mechanism; optimised unary encoding (OUE) frequency oracle | Implemented | Fig. 5 |
| **3.3** Piecewise mechanism | Piecewise mechanism for numeric features before training | Experimental | Tab. 5 |
| **3.4** LDP Naive Bayes | Feature-sampling LDP-NB over public clinical bins | Experimental | Tab. 5, Fig. 5 |
| **4.1** Privacy in data processing & mining | Defences grouped by pipeline stage | Analytical | Sec. 2.2 |
| **4.2** Modifying the input | Additive random perturbation ($\sigma$ = 0.25, 0.5) | Experimental | Tab. 6 |
| **4.3** Publishing data | Mondrian multidimensional k-anonymity on Age, Pregnancies, BMI | Implemented | `m4_ppdm.py`, Tab. 6 |
| **4.4** Beyond k-anonymity | Distinct l-diversity constraint; homogeneity exposure; t-closeness | Implemented | Tab. 7, Fig. 6 |

| Syllabus topic | How it is used in this work | Treatment | Evidence |
|---|---|---|---|
| **4.5** Modifying the mining output | API returns rounded scores or labels only | Experimental | Tab. 6, Fig. 6 |
| **5.1** Compressive privacy | Clients release k-dim projection $z = W^T x$, not x | Implemented | `m5_compressive.py` |
| **5.2** CP mechanisms | PCA (unsupervised), DCA (utility-supervised), DUCA (utility vs privacy label) | Implemented | Fig. 7 |
| **5.3** CP for ML applications | Diabetes classifier trained on z; MIA, attribute inference and reconstruction attacks on z | Experimental | Tab. 8 |
| **5.4** Accuracy of the utility task | Utility accuracy vs k for each projection | Experimental | Fig. 7 |

## 2.2 How the modules fit together, and where they do not apply directly

The modules protect different parts of the pipeline and assume different trust models. **Input-side** techniques (LDP, input noise, k-anonymity, compressive projection) change what the data holder receives or releases. **Training-side** techniques (regularisation, DP-GD) limit how much the model memorises. **Output-side** techniques (rounding, label-only) limit what the API reveals. Measuring all of them with one attack makes this difference visible: only the formal methods (DP, LDP) bound leakage for *every* attacker, while the others reduce leakage only against the attacks that were tried.

**Where a module is not directly applicable, it is handled as follows.**

- **Module 1.4 / 4.1** are overview topics. They have no algorithm of their own and are covered analytically through the input/training/output grouping above.
- **Module 2.4 (unsupervised DP).** Membership inference here targets a supervised classifier, so DP clustering cannot be attacked in the same harness. It is shown as a controlled demonstration (DP k-means cost vs ε, Fig. 4) instead.
- **Module 3 on the second dataset.** WDBC has 30 continuous features and no public clinical bounds. Splitting a per-patient budget over 31 attributes leaves under 0.3 per attribute even at $\varepsilon = 8$, so LDP training is not meaningful at $n \approx 142$. LDP is therefore evaluated on Pima only, plus a bootstrap simulation that asks how many users would be needed.
- **Module 4 (k-anonymity) on the second dataset.** WDBC contains only cell-nucleus measurements and no demographic quasi-identifiers, so there is nothing to generalise. k-anonymity is evaluated on Pima (QIs: Age, Pregnancies, BMI). Input noise and output restriction still apply to both datasets.
- **Module 5 vs the MIA threat.** Compressive privacy protects the *released data* rather than the trained model, so its primary metrics are attribute inference and reconstruction. MIA against a model trained on z is reported as well, but a low-dimensional input does not by itself stop an over-parameterised model from memorising. DUCA needs a privacy label and is therefore Pima-only (Age >= 35).

# 3. Dataset Finalisation and Individual Implementation

## 3.1 Dataset provenance record

| Item | Pima Indians Diabetes (primary) | Breast Cancer Wisconsin Diagnostic (replication) |
|---|---|---|
| Source / URL | Public mirror: raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv (mirror of the UCI ML Repository version) | Copy bundled with scikit-learn 1.6.1 (`sklearn.datasets.load_breast_cancer`); original: UCI ML Repository, DOI 10.24432/C5DW2B |
| Original author / publisher | National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK); Smith et al. (1988); donated to UCI by V. Sigillito (JHU APL), 1990 | W. H. Wolberg, W. N. Street, O. L. Mangasarian, University of Wisconsin (1995) |
| Collection method (original) | Clinical measurements (2-h OGTT glucose, BP, skinfold, insulin, BMI, pedigree function, age) for female patients >= 21 years of Pima heritage; outcome = diabetes within 5 years | 30 features computed from digitised images of fine-needle aspirates of breast masses; diagnosis malignant/benign |
| Collection date (this work) | Downloaded 2026-10-23 (UTC) via HTTPS | Loaded locally 2026-10-23 from the installed package |
| Licence / usage terms | The mirror has no licence file (GitHub API reports none). The UCI page is no longer online; a widely used redistribution (Kaggle 'uciml') lists CC0. **Exact licence of the mirror: unknown.** Used for non-commercial academic benchmarking only. | CC BY 4.0 (UCI). |
| Version / hash | Mirror commit 357ed4a (10 Mar 2018); SHA-256 6bfe5d0f379d17a0e0819b996407e3c0... | SHA-256 fed3eb72d0575ef6192293f5093c6e80... |
| Original size | 768 rows x 9 columns; class 0/1 = 500/268 | 569 rows x 30 features; malignant/benign = 212/357 |
| Final size | 768 rows x 10 features (8 scaled + 2 missing indicators) + label; no rows dropped | 569 x 30 scaled + label; no rows dropped |
| Access restrictions & safeguards | Public, de-identified. No linkage with other data, no re-identification attempted; MIA run only on local models | Public, de-identified, no demographics |
| Known limitations | Small (768); one population and sex only; zeros encode missing values; ethical concerns about secondary use of data from an Indigenous community (Sec. 6) | Small (569); derived image features; not representative of clinical populations |

## 3.2 Preprocessing log (Pima)

| Step | Operation | Effect |
|---|---|---|
| 1 | Verify SHA-256 of the raw file on every load | Pipeline aborts if the file changes |
| 2 | Zero -> missing for Glucose, BloodPressure, SkinThickness, Insulin, BMI | Missing: Glucose 5, BP 35, Skin 227, Insulin 374, BMI 11 |
| 3 | Clip each feature to fixed public bounds (e.g. Glucose 40-200, Age 21-81) and scale to [-1, 1] | Data-independent: costs no privacy budget, so DP/LDP guarantees remain valid |
| 4 | Missing values -> centre of the public range (0 after scaling) | No data-derived imputation |
| 5 | Add missing indicators for SkinThickness and Insulin | Keeps the information that a test was not performed; 10 model features in total |
| 6 | Discretise with public clinical cut points (e.g. BMI 25/30/35; Age 30/40/50) for LDP-NB | 4-5 bins per feature; written to data/processed/pima_discretised.csv |
| 7 | Stratified split per seed (seeds 0-9): 192 members / 192 non-members / 384 shadow pool | Non-members double as the utility test set |

For WDBC, features are scaled to [-1, 1] using the minimum and maximum values published in the dataset documentation (treated as public metadata), and the label is recoded so that malignant = 1. Processed files and their hashes are listed in `results/tables/provenance.json`.

## 3.3 Implementation

| File | Content (written for this IA) |
|---|---|
| `src/common.py` | Loading with hash check, public-bounds scaling, stratified splits, utility metrics, NumPy MLP with per-example gradients |
| `src/attacks.py` | Shadow-model attack, loss-threshold attack, API modes (full / rounded / label), attack metrics including advantage |
| `src/models.py` | Model matrix A-E (complexity x regularisation) |
| `src/m2_dp.py` | DP-GD, zCDP accountant ($\sigma$ <-> $\varepsilon$), DP-advantage bound, DPLloyd k-means |
| `src/m3_ldp.py` | Laplace / Duchi / Piecewise / Hybrid mechanisms, randomised response, OUE, LDP Naive Bayes |
| `src/m4_ppdm.py` | Mondrian k-anonymity with l-diversity, NCP / homogeneity / t-closeness, input noise |
| `src/m5_compressive.py` | PCA, DCA, DUCA projections; projected-model wrapper |
| `src/run_experiments.py` | All configurations x 10 seeds in parallel; standalone experiments; provenance |
| `src/make_figures.py` | Summary tables, paired statistics, all figures |

| File | Content (written for this IA) |
|---|---|
| `src/build_report.py`, `src/build_manuscript.py` | This report and the IEEE-style manuscript |

### *Key algorithms*

**DP-GD and accounting.** For T = 200 full-batch steps, each per-example gradient is clipped to L2 norm C = 1, the clipped gradients are summed, and $N(0, \sigma^2C^2I)$ noise is added. One step is $1/(2\sigma^2)$-zCDP under add/remove adjacency, so T steps give $\rho = T/(2\sigma^2)$ and $\varepsilon = \rho + 2\sqrt{(\rho \ln(1/\delta))}$ with $\delta = 10^{-5}$. Inverting this gives the noise used: $\sigma = 69.3$ for $\varepsilon = 1$, 18.3 for $\varepsilon = 4$ and 9.8 for $\varepsilon = 8$. Without subsampling there is no privacy amplification, so the stated $\varepsilon$ is conservative.

**Piecewise mechanism.** For $x \in [-1, 1]$ and budget $\varepsilon$, with $C = (e^{(\varepsilon/2)}+1)/(e^{(\varepsilon/2)}-1)$, the report is drawn uniformly from $[l(x), r(x)]$ with probability $e^{(\varepsilon/2)}/(e^{(\varepsilon/2)}+1)$, and otherwise uniformly from the rest of $[-C, C]$. Here $l(x) = (C+1)x/2 - (C-1)/2$ and $r(x) = l(x) + C - 1$. The output is unbiased. Each record's 10 features and its label share $\varepsilon$ equally (sequential composition).

**LDP Naive Bayes.** Each patient samples one feature j and reports the joint value (class, bin_j) through OUE ($p = 1/2$, $q = 1/(e^{\varepsilon}+1)$) with the full $\varepsilon$. The aggregator debiases the counts, clips negatives and forms the class prior and class-conditional tables.

**Mondrian with l-diversity.** The partitioner recursively median-splits the QI with the widest range and accepts a split only if both halves have >= k records and >= l distinct outcomes. QI values are then replaced by their equivalence-class mean.

**DCA / DUCA.** DCA solves the generalised eigenproblem $(S_B + \rho'I)w = \lambda(S_W + S_B + \rho I)w$ for the utility label. DUCA uses $(S_B^u + \rho'I)w = \lambda(\gamma S_B^p + S_W^u + \rho I)w$, which penalises directions that separate the privacy attribute ($\gamma = 5$, $\rho = 10^{-2}$, $\rho' = 10^{-3}$).

## 3.4 Execution instructions

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.lock.txt
.venv/bin/python src/run_experiments.py # add --quick for a 2-seed smoke test
.venv/bin/python src/make_figures.py && .venv/bin/python src/build_report.py
```

The full run performs 570 target/attack evaluations (9 models each) and took roughly 0.3 h wall-clock on 8 worker processes (10-core laptop CPU).

## 3.5 Individual-contribution statement

This IA-2 is my individual work (Anjali Ashtankar, 16014223012). Building on my IA-1 foundation document, I contributed:

- **Problem refinement:** reframed the IA-1 attack as a common yardstick for all five modules and wrote RQ1-RQ6.
- **Dataset work:** sourced and hashed both datasets; designed the data-independent preprocessing (public clinical bounds, zero-as-missing handling, missing indicators, clinical bins) needed for valid DP/LDP claims.
- **Implementation:** wrote the attack harness, DP-GD with its zCDP accountant, the four LDP mechanisms and LDP-NB, Mondrian with l-diversity, and DCA/DUCA from first principles in NumPy (scikit-learn is used only for standard non-private models and metrics).
- **Experimental design:** Kerckhoffs-style shadow models, three API modes, 10-seed paired design, advantage noise floor, DP-bound comparison, bootstrap user-scaling simulation, second-dataset replication.
- **Analysis and documentation:** statistical testing, figures, this report, and the manuscript draft.

# 4. Evaluation, Comparison and Critical Analysis

## 4.1 Metrics

- **Utility:** test accuracy, F1 and ROC-AUC on the held-out non-members; generalisation gap = train acc - test acc.
- **Attack ROC-AUC:** threshold-free probability that a random member scores above a random non-member (0.5 = random guessing). *Worst-case AUC* is the larger of the shadow and loss attacks.
- **Attack accuracy / precision / recall / F1** at the threshold calibrated on the shadow models (balanced 192 vs 192 evaluation).
- **Advantage** = TPR - FPR (Yeom et al.). *Max advantage* is the best threshold in hindsight; it is compared with the DP bound (e^ε - 1 + 2δ)/(e^ε + 1).
- **Statistics:** mean ± SD over 10 seeds; paired differences vs the baseline with 95% t-intervals and two-sided Wilcoxon signed-rank p-values.

## 4.2 Module 1 - model complexity and overfitting (RQ1)

***Table 3.*** *Model matrix on Pima (mean ± SD where shown; attack metrics are for the shadow attack).*

| Model | Train acc | Test acc | Gap | Shadow AUC | Loss AUC | Prec | Rec | F1 | Adv |
|---|---|---|---|---|---|---|---|---|---|
| A: LogReg (L2) | 0.773 | 0.773 ± 0.017 | 0.001 | 0.511 ± 0.023 | 0.504 | 0.501 | 0.674 | 0.572 | 0.006 |
| B: MLP-32 | 0.858 | 0.762 ± 0.019 | 0.096 | 0.571 ± 0.042 | 0.548 | 0.529 | 0.820 | 0.642 | 0.092 |
| C: MLP-128x2 **(baseline)** | 1.000 | 0.719 ± 0.026 | 0.281 | 0.765 ± 0.018 | 0.631 | 0.640 | 0.883 | 0.735 | 0.376 |
| D: MLP-128x2 + L2 + early stop | 0.721 | 0.711 ± 0.071 | 0.010 | 0.494 ± 0.018 | 0.505 | 0.493 | 0.566 | 0.516 | -0.014 |
| E: Random forest (deep) | 1.000 | 0.759 ± 0.017 | 0.241 | 0.798 ± 0.024 | 0.766 | 0.678 | 0.856 | 0.756 | 0.448 |

Leakage follows the generalisation gap closely (Fig. 2). The unregularised MLP-128x2 (baseline C) memorises its 192 training records (gap 0.281), and the shadow attack separates members from non-members with AUC 0.765. The deep random forest leaks even more (AUC 0.798) despite better test accuracy, so **high accuracy does not imply low leakage**. Adding L2 and early stopping to the same architecture (model D) brings the AUC down to 0.494 with test accuracy 71.1% versus 71.9% (difference not significant, p = 0.846). On this small table the simplest model, logistic regression (A), is both the most accurate (77.3%) and essentially leak-free (AUC 0.511): capacity beyond what the data supports buys memorisation, not accuracy. Regularisation and model choice are therefore an effective first line of defence, but they come with no guarantee.

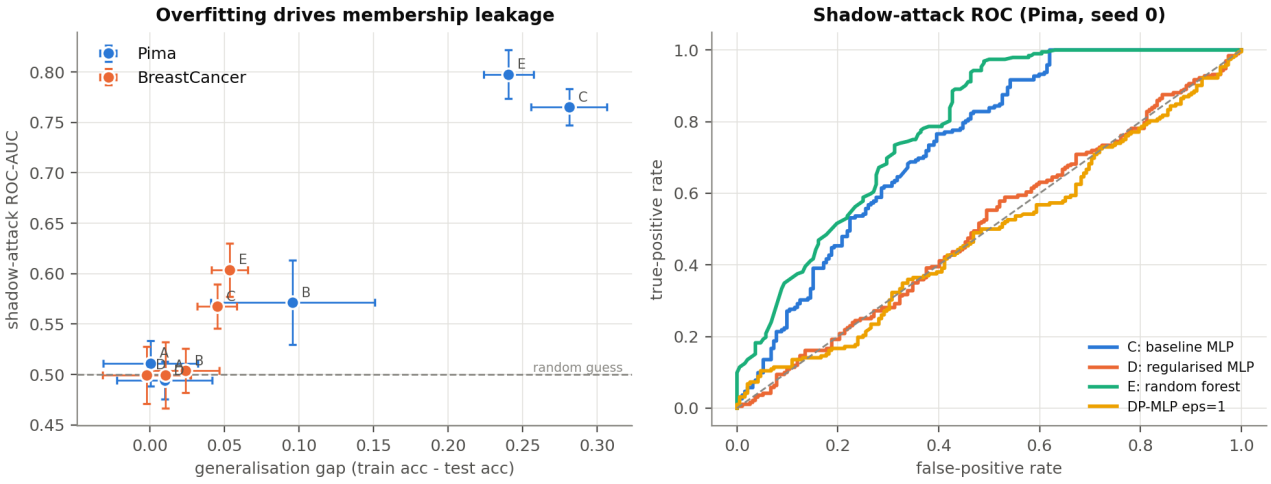



*Figure 2. Left: attack AUC vs generalisation gap for models A-E on both datasets (error bars = SD over seeds). Right: shadow-attack ROC curves on one Pima split.*

## 4.3 Module 2 - central differential privacy (RQ2)

***Table 4.*** *DP gradient descent on Pima ($\delta = 10^{-5}$, $C = 1$, $T = 200$).*

| Configuration | σ | Test acc | Gap | Shadow AUC | Loss AUC | Max adv. | DP bound |
|---|---|---|---|---|---|---|---|
| DP-GD LogReg ε = 0.5 | 137.2 | 0.623 ± 0.108 | -0.029 | 0.494 ± 0.029 | 0.487 | 0.058 | 0.245 |
| DP-GD LogReg ε = 1 | 69.3 | 0.686 ± 0.055 | 0.010 | 0.493 ± 0.031 | 0.500 | 0.060 | 0.462 |
| DP-GD LogReg ε = 2 | 35.3 | 0.743 ± 0.036 | -0.002 | 0.504 ± 0.028 | 0.500 | 0.059 | 0.762 |
| DP-GD LogReg ε = 4 | 18.3 | 0.758 ± 0.028 | 0.010 | 0.516 ± 0.033 | 0.502 | 0.077 | 0.964 |
| DP-GD LogReg ε = 8 | 9.8 | 0.768 ± 0.014 | 0.007 | 0.512 ± 0.032 | 0.502 | 0.066 | 0.999 |
| DP-GD LogReg ε = ∞ (non-private, T = 200) | 0 | 0.774 ± 0.018 | -0.003 | 0.513 ± 0.029 | 0.505 | 0.077 | – |
| DP-GD MLP-128 ε = 0.5 | 137.2 | 0.557 ± 0.090 | 0.002 | 0.519 ± 0.038 | 0.508 | 0.076 | 0.245 |
| DP-GD MLP-128 ε = 1 | 69.3 | 0.594 ± 0.063 | 0.015 | 0.498 ± 0.044 | 0.505 | 0.064 | 0.462 |
| DP-GD MLP-128 ε = 2 | 35.3 | 0.687 ± 0.034 | 0.020 | 0.500 ± 0.028 | 0.510 | 0.066 | 0.762 |
| DP-GD MLP-128 ε = 4 | 18.3 | 0.747 ± 0.015 | 0.020 | 0.510 ± 0.017 | 0.511 | 0.071 | 0.964 |
| DP-GD MLP-128 ε = 8 | 9.8 | 0.758 ± 0.012 | 0.012 | 0.518 ± 0.018 | 0.509 | 0.067 | 0.999 |
| DP-GD MLP-128 ε = ∞ (non-private, T = 200) | 0 | 0.767 ± 0.015 | 0.029 | 0.517 ± 0.037 | 0.515 | 0.074 | – |
| NumPy MLP-128, long training (T = 2000, non-private) | 0 | 0.700 ± 0.028 | 0.298 | 0.741 ± 0.029 | 0.615 | 0.376 | – |

Utility rises with ε as expected (Fig. 3). At ε = 4 the DP-MLP reaches 74.7% accuracy (DP logistic regression 75.8%), compared with 71.9% for the non-private baseline, while the worst-case attack AUC stays at 0.511. Below ε = 1 the noise dominates: the DP-MLP falls to 59.4% at ε = 1. The **ablation** separates the two causes of the improvement. The same NumPy MLP trained without noise for 2000 steps leaks (AUC 0.741), but trained for only 200 steps it already leaks little (AUC 0.517). Most of the *empirical* reduction against these attacks therefore comes from the short training schedule that DP forces. What DP adds is a **provable bound that holds for every attacker**, including stronger ones than those tested here.

Across all ε, the measured maximum advantage stays inside the random-score noise floor (grey band, Fig. 3 right) and far below the theoretical bound. This is consistent with DP, and it also shows that the bound is loose for average-case attackers at moderate ε. Fig. 4 shows the two DP properties used: sequential composition (ε grows with the number of noisy steps for fixed σ) and an unsupervised application, DP k-means. Its clustering cost approaches the non-private cost as ε grows (SSE ratio 1.06 at ε = 8, 3.27 at ε = 0.5).

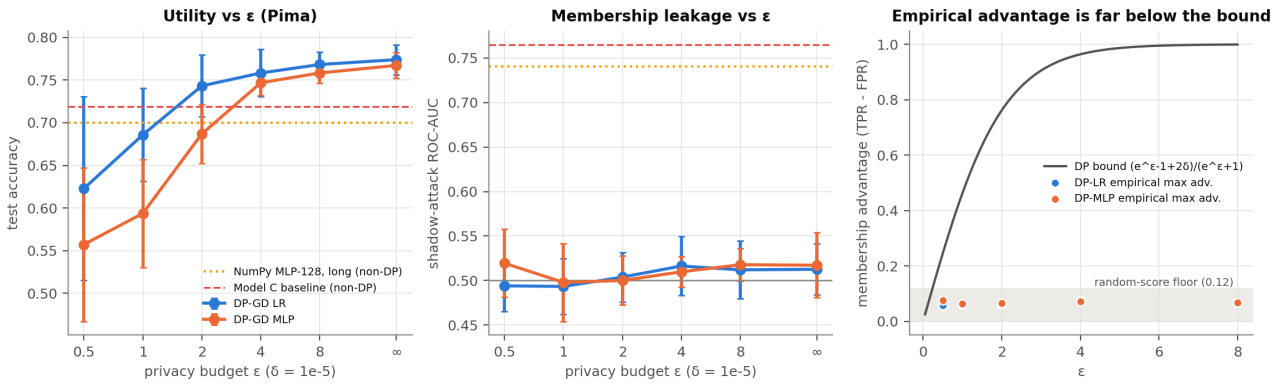



*Figure 3. DP-GD on Pima: utility (left) and attack AUC (centre) vs ε; right: empirical maximum advantage vs the (ε, δ)-DP bound, with the 95th-percentile advantage of a random attacker shown as a grey band.*

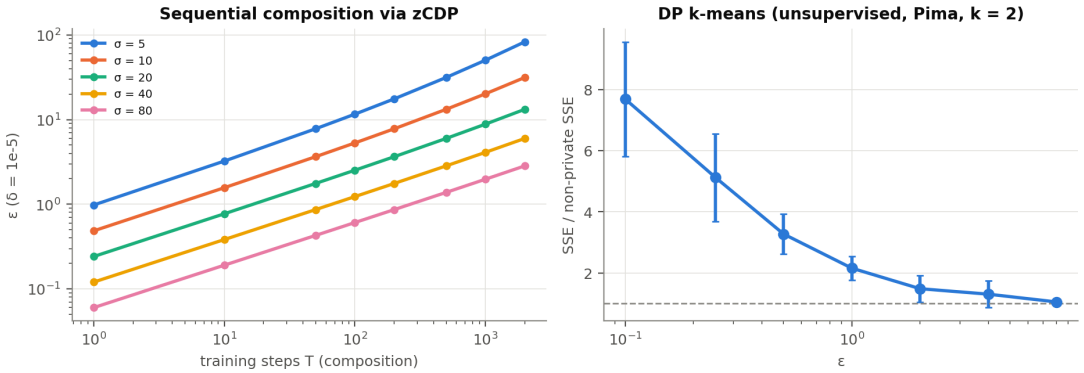



*Figure 4. Left: zCDP composition - ε after T Gaussian steps for several noise multipliers. Right: DP k-means (DPLloyd, 5 iterations) cost relative to non-private k-means (30 trials).*

## 4.4 Module 3 - local differential privacy (RQ3)

***Table 5.*** *Models trained on locally privatised Pima records (n = 192 per model; ε is the per-patient budget).*

| Configuration | Test acc | Test F1 | Shadow AUC | Loss AUC | Max adv. |
|---|---|---|---|---|---|
| LDP-Piecewise LogReg ε = 1 | 0.548 ± 0.140 | 0.167 | 0.497 ± 0.021 | 0.509 | 0.069 |
| LDP-Piecewise LogReg ε = 4 | 0.619 ± 0.070 | 0.035 | 0.508 ± 0.036 | 0.509 | 0.074 |
| LDP-Piecewise LogReg ε = 8 | 0.644 ± 0.032 | 0.071 | 0.490 ± 0.028 | 0.502 | 0.060 |
| LDP-Piecewise LogReg ε = 16 | 0.657 ± 0.017 | 0.111 | 0.497 ± 0.018 | 0.500 | 0.056 |
| LDP Naive Bayes ε = 1 | 0.578 ± 0.086 | 0.370 | 0.497 ± 0.017 | 0.505 | 0.059 |
| LDP Naive Bayes ε = 2 | 0.609 ± 0.047 | 0.424 | 0.513 ± 0.032 | 0.496 | 0.076 |
| LDP Naive Bayes ε = 4 | 0.668 ± 0.035 | 0.410 | 0.514 ± 0.022 | 0.501 | 0.070 |
| LDP Naive Bayes ε = 8 | 0.691 ± 0.036 | 0.480 | 0.506 ± 0.031 | 0.506 | 0.065 |
| LDP Naive Bayes ε = ∞ (non-private) | 0.752 ± 0.020 | 0.648 | 0.525 ± 0.031 | 0.529 | 0.087 |

The correctness check (Fig. 5, left) confirms that all four mechanisms are unbiased and that their empirical per-report variance matches theory. Duchi has lower variance at small ε (≤ 0.5) and Piecewise wins from ε = 1 upward, which reproduces the crossover reported by Wang et al. (2019). Hybrid matches Duchi at small ε and lies between the two at larger ε here, because it minimises *worst-case* variance while this test input is skewed towards the ends of the range. LDP gives the strongest trust model, since the hospital never sees raw values, and every LDP-trained model shows attack AUC close to 0.5. The cost is utility at this data size. Piecewise perturbation of all 11 attributes reaches only 65.7% at ε = 16, because each attribute gets ε/11. LDP Naive Bayes, where each user reports one sampled feature with the full ε, reaches 66.8% at ε = 4 versus 75.2% non-privately. The bootstrap simulation (Fig. 5, centre) shows what scale is needed: with 50,000 reporting users LDP-NB at ε = 1 reaches 74.1%, compared with 60.2% using the 576 real records. **LDP is the right tool for population-scale collection, but not for a single small clinic.**

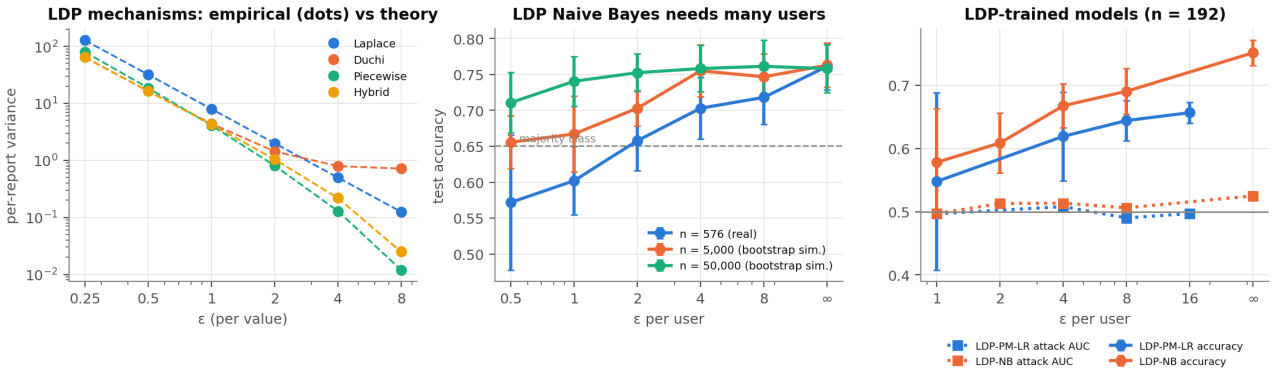



*Figure 5. Left: per-report variance of the LDP mechanisms (dots = empirical, dashed = theory). Centre: LDP Naive Bayes accuracy vs ε for the real 576 records and bootstrap simulations of larger user bases. Right: accuracy and attack AUC of LDP-trained models.*

## 4.5 Module 4 - privacy-preserving data mining (RQ4)

***Table 6.*** *PPDM defences on Pima. NCP = normalised certainty penalty (information loss) of the anonymised training set; Homog. = share of records in equivalence classes with a single outcome.*

| Configuration | Test acc | Gap | Shadow AUC | Loss/label AUC | Adv | NCP | Homog. |
|---|---|---|---|---|---|---|---|
| Model C, full probabilities | 0.719 ± 0.026 | 0.281 | 0.765 ± 0.018 | 0.631 | 0.376 | – | – |
| Model C, rounded API (4.5) | 0.719 ± 0.026 | 0.281 | 0.707 ± 0.013 | 0.681 | 0.362 | – | – |
| Model C, label-only API (4.5) | 0.719 ± 0.026 | 0.281 | 0.666 ± 0.022 | 0.641 | 0.281 | – | – |
| Mondrian k=5 | 0.702 ± 0.031 | 0.254 | 0.675 ± 0.024 | 0.617 | 0.267 | 0.137 | 0.199 |
| Mondrian k=10 | 0.709 ± 0.019 | 0.220 | 0.647 ± 0.016 | 0.596 | 0.217 | 0.215 | 0.086 |
| Mondrian k=25 | 0.694 ± 0.022 | 0.160 | 0.598 ± 0.015 | 0.576 | 0.151 | 0.389 | 0.029 |
| Mondrian k=10, l=2 | 0.714 ± 0.022 | 0.217 | 0.640 ± 0.018 | 0.595 | 0.218 | 0.225 | 0.000 |
| Input noise σ=0.25 | 0.701 ± 0.036 | 0.103 | 0.565 ± 0.031 | 0.548 | 0.085 | – | – |
| Input noise σ=0.5 | 0.689 ± 0.046 | 0.043 | 0.521 ± 0.032 | 0.521 | 0.022 | – | – |

***Table 7.*** *Anonymity analysis of the full Pima table (QIs: Age, Pregnancies, BMI; sensitive attribute: diabetes outcome).*

| k | l | Classes | Avg size | NCP | Homogeneous | t (max TV dist.) |
|---|---|---|---|---|---|---|
| 2 | 1 | 321 | 2.4 | 0.033 | 54.2% | 0.651 |
| 5 | 1 | 117 | 6.6 | 0.085 | 22.9% | 0.651 |
| 10 | 1 | 56 | 13.7 | 0.131 | 15.1% | 0.589 |
| 25 | 1 | 22 | 34.9 | 0.225 | 13.8% | 0.358 |
| 50 | 1 | 10 | 76.8 | 0.323 | 9.2% | 0.349 |
| 2 | 2 | 164 | 4.7 | 0.078 | 0.0% | 0.589 |
| 5 | 2 | 94 | 8.2 | 0.104 | 0.0% | 0.589 |
| 10 | 2 | 50 | 15.4 | 0.144 | 0.0% | 0.589 |
| 25 | 2 | 21 | 36.6 | 0.229 | 0.0% | 0.358 |
| 50 | 2 | 10 | 76.8 | 0.325 | 0.0% | 0.338 |

**Output restriction (4.5)** helps less than expected. Rounding scores lowers the worst-case AUC to 0.707, and returning labels only lowers it to 0.666 (baseline 0.765). An overfit model is simply *correct* far more often on its members, and a label-only attacker exploits that, as Choquette-Choo et al. (2021) predict. **k-anonymity (4.3)** reduces the attack AUC from 0.765 to 0.675 (k = 5) and 0.598 (k = 25) as generalisation blurs the training records, but the model still overfits the remaining features. It also has a privacy flaw of its own: at k = 5, 22.9% of records (k = 10: 15.1%) sit in classes whose members all share the same diagnosis, so anyone who knows a patient's age, pregnancies and BMI learns the outcome (a homogeneity attack). **Adding l = 2 diversity (4.4)** drives this exposure to 0% for every k, at a small extra information loss (NCP 0.131 -> 0.144 at k = 10). **Input randomisation (4.2)** with σ = 0.5 acts as a strong regulariser (AUC 0.521, accuracy 68.9%) but carries no formal guarantee.

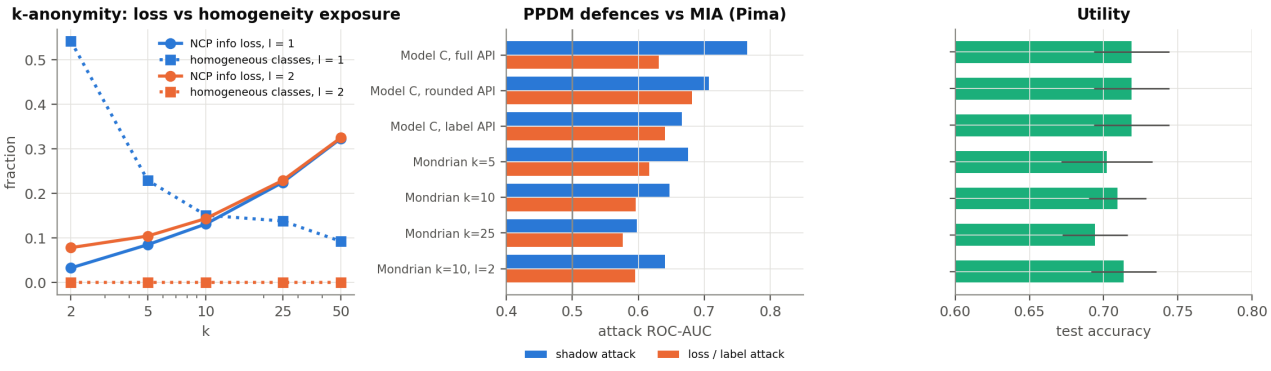



*Figure 6. Left: information loss and homogeneity exposure vs k, with and without l-diversity. Centre/right: attack AUC and utility for the PPDM defences on Pima.*

## 4.6 Module 5 - compressive privacy (RQ5, syllabus 5.4)

***Table 8.*** *Compressive privacy on Pima. Attribute inference: random forest trained on the attacker's pool to predict Age >= 35 from z. Reconstruction: linear regression from z to x (NRMSE 1 = no better than the mean).*

| Projection | Utility acc (MLP) | Shadow AUC | Loss AUC | Age-inference bal. acc | Recon. NRMSE |
|---|---|---|---|---|---|
| PCA k=2 | 0.699 ± 0.030 | 0.538 ± 0.032 | 0.528 | 0.855 | 0.509 |
| PCA k=4 | 0.705 ± 0.020 | 0.695 ± 0.022 | 0.597 | 0.902 | 0.333 |
| DCA k=1 | 0.745 ± 0.027 | 0.537 ± 0.024 | 0.535 | 0.623 | 0.957 |
| DCA k=2 | 0.722 ± 0.033 | 0.651 ± 0.035 | 0.577 | 0.653 | 0.956 |
| DCA k=4 | 0.691 ± 0.028 | 0.783 ± 0.022 | 0.626 | 0.674 | 0.950 |
| DUCA k=2 | 0.695 ± 0.026 | 0.669 ± 0.026 | 0.595 | 0.536 | 0.967 |
| Raw features (reference, logistic regression) | 0.773 | – | – | 1.000 | 0.000 |

Fig. 7 shows that the choice of projection matters more than its dimension. **PCA** keeps the directions of largest variance, so it stays reconstructable (NRMSE 0.51 at k = 2) and leaks age (balanced accuracy 0.80 even at k = 1), while its utility at small k is poor (65.1% at k = 1). **DCA** puts the diabetes signal into its first component: one number per patient gives 77.2% accuracy (raw data: 77.3%) while age inference falls to 0.62 and reconstruction is almost impossible (NRMSE 0.96 at k = 2). **DUCA** additionally penalises the age direction and pushes age inference close to chance (0.51 at k = 1) for a small utility cost (76.7%). On the MIA axis, an over-parameterised MLP on z still memorises once z has enough dimensions (DCA/DUCA k >= 2, PCA k >= 4). With k = 1 (DCA) the worst-case AUC is 0.537 at 74.5% accuracy. So compressive privacy protects the released data, and has to be combined with training-side measures to protect the model.

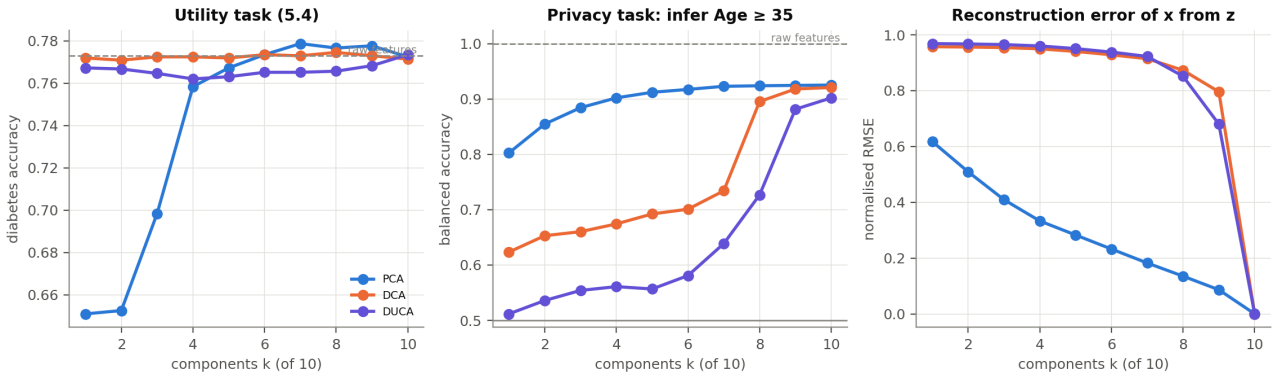



*Figure 7. Compressive privacy on Pima vs number of components k (mean over seeds): utility-task accuracy (logistic regression on z), privacy-task balanced accuracy, and reconstruction error.*

## 4.7 Cross-module comparison (RQ6)

***Table 9.*** *Selected configurations vs baseline C on Pima (paired over 10 seeds; Wilcoxon signed-rank p-values; Δ = configuration - baseline).*

| Mod. | Configuration | Test acc | Worst AUC | Δ shadow AUC [95% CI] | p | Δ acc [95% CI] | p | Formal guarantee |
|---|---|---|---|---|---|---|---|---|
| M1 | Baseline C (reference) | 0.719 | 0.765 | – | – | – | – | none |
| M1 | D: MLP-128x2 + L2 + early stop | 0.711 | 0.505 | -0.271 [±0.022] | 0.002 | -0.008 [±0.053] | 0.846 | none |
| M4 | C: MLP-128x2, label API | 0.719 | 0.666 | -0.099 [±0.019] | 0.002 | +0.000 [±0.000] | 1.000 | none |
| M2 | DP-GD LogReg ε = 4 | 0.758 | 0.516 | -0.249 [±0.022] | 0.002 | +0.040 [±0.012] | 0.002 | (ε, δ)-DP |
| M2 | DP-GD MLP-128 ε = 4 | 0.747 | 0.511 | -0.255 [±0.017] | 0.002 | +0.028 [±0.017] | 0.020 | (ε, δ)-DP |
| M2 | DP-GD MLP-128 ε = 1 | 0.594 | 0.505 | -0.267 [±0.031] | 0.002 | -0.125 [±0.053] | 0.002 | (ε, δ)-DP |
| M3 | LDP Naive Bayes ε = 4 | 0.668 | 0.514 | -0.252 [±0.016] | 0.002 | -0.051 [±0.036] | 0.021 | ε-LDP per record |
| M3 | LDP-Piecewise LogReg ε = 16 | 0.657 | 0.500 | -0.268 [±0.017] | 0.002 | -0.062 [±0.022] | 0.002 | ε-LDP per record |
| M4 | Mondrian k=25 | 0.694 | 0.598 | -0.167 [±0.018] | 0.002 | -0.024 [±0.027] | 0.084 | k-anon. / l-div. (syntactic) |
| M4 | Mondrian k=10, l=2 | 0.714 | 0.640 | -0.125 [±0.014] | 0.002 | -0.005 [±0.017] | 0.557 | k-anon. / l-div. (syntactic) |
| M4 | Input noise σ=0.5 | 0.689 | 0.521 | -0.244 [±0.028] | 0.002 | -0.030 [±0.034] | 0.131 | none |
| M5 | DCA k=1 | 0.745 | 0.537 | -0.228 [±0.026] | 0.002 | +0.026 [±0.024] | 0.051 | none (info-theoretic intuition) |
| M5 | DUCA k=2 | 0.695 | 0.669 | -0.097 [±0.016] | 0.002 | -0.023 [±0.021] | 0.037 | none (info-theoretic intuition) |

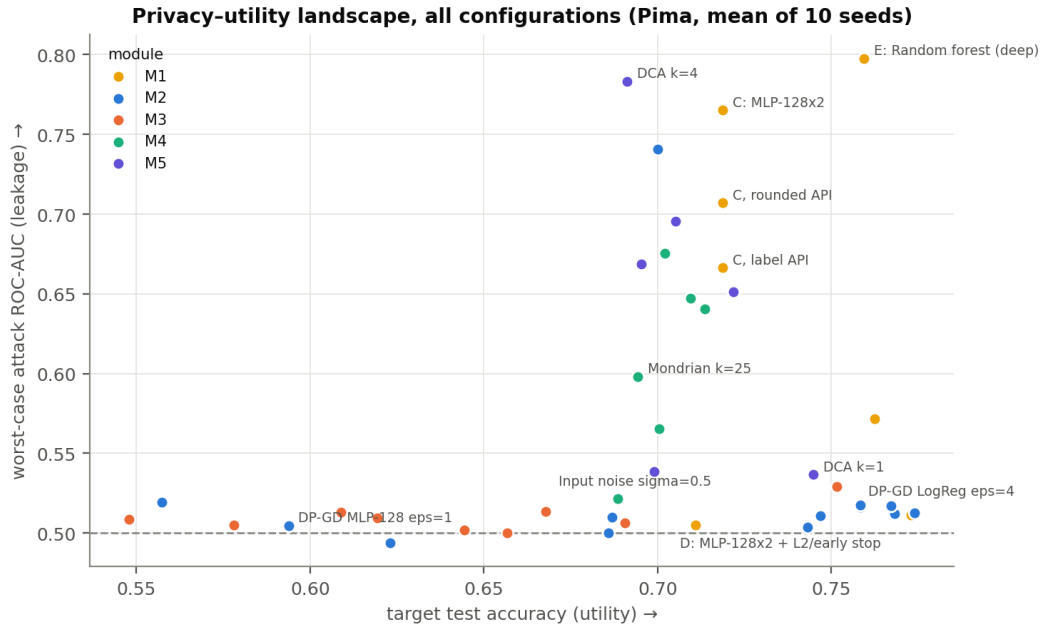



*Figure 8. Privacy-utility landscape of every configuration on Pima. Lower-right is better (high accuracy, attack AUC near 0.5).*

Fig. 8 brings all modules together. The lower-right region (high accuracy, attack AUC near 0.5) is occupied by the **simple logistic regression (A)**, **DP-GD at ε = 4-8** and **DCA with k = 1**. The regularised MLP (D) and input noise also reach chance-level leakage at slightly lower accuracy, and the LDP models reach it at a larger accuracy cost. Of these, only DP-GD and LDP come with a formal guarantee. Output restriction and k-anonymity leave the overfit model in the upper half. For this use case the recommended configuration is **DP-GD logistic regression at ε ≈ 4**. It reaches 75.8% accuracy, only 1.6 points below its non-private counterpart and 4.0 points above the leaky baseline, with attack AUC ≈ 0.52, and its guarantee also covers attackers stronger than the ones tested. If records are released rather than models, DCA/DUCA projections are the preferred release format.

### *Results recorded for the IA-1 table (Section 13 of IA-1)*

| Model | Test accuracy | MIA accuracy | MIA precision | MIA recall | MIA F1 | MIA ROC-AUC |
|---|---|---|---|---|---|---|
| Baseline (C) | 0.719 | 0.688 | 0.640 | 0.883 | 0.735 | 0.765 |

| Model | Test accuracy | MIA accuracy | MIA precision | MIA recall | MIA F1 | MIA ROC-AUC |
|---|---|---|---|---|---|---|
| Regularised (D) | 0.711 | 0.493 | 0.493 | 0.566 | 0.516 | 0.494 |
| DP model (DP-GD MLP, ε = 4) | 0.747 | 0.499 | 0.500 | 0.601 | 0.542 | 0.510 |

## 4.8 Replication on Breast Cancer Wisconsin

***Table 10.*** *All configurations on Breast Cancer Wisconsin (Diagnostic).*

| Configuration | Test acc | Gap | Shadow AUC | Loss AUC | Recon. NRMSE |
|---|---|---|---|---|---|
| A: LogReg (L2) | 0.968 ± 0.014 | 0.011 | 0.499 ± 0.033 | 0.500 | – |
| B: MLP-32 | 0.963 ± 0.015 | 0.024 | 0.504 ± 0.022 | 0.509 | – |
| C: MLP-128x2 (baseline) | 0.955 ± 0.013 | 0.045 | 0.568 ± 0.022 | 0.517 | – |
| C: MLP-128x2, rounded API | 0.955 ± 0.013 | 0.045 | 0.537 ± 0.010 | 0.532 | – |
| C: MLP-128x2, label API | 0.955 ± 0.013 | 0.045 | 0.529 ± 0.011 | 0.523 | – |
| D: MLP-128x2 + L2 + early stop | 0.918 ± 0.038 | -0.002 | 0.499 ± 0.028 | 0.506 | – |
| E: Random forest (deep) | 0.946 ± 0.012 | 0.054 | 0.604 ± 0.026 | 0.608 | – |
| NumPy MLP-128, long training (non-private) | 0.951 ± 0.018 | 0.049 | 0.546 ± 0.021 | 0.516 | – |
| DP-GD LogReg ε = 1 | 0.832 ± 0.094 | 0.004 | 0.502 ± 0.030 | 0.509 | – |
| DP-GD MLP-128 ε = 1 | 0.793 ± 0.074 | -0.013 | 0.502 ± 0.032 | 0.490 | – |
| DP-GD LogReg ε = 4 | 0.940 ± 0.027 | 0.011 | 0.507 ± 0.023 | 0.511 | – |
| DP-GD MLP-128 ε = 4 | 0.932 ± 0.017 | 0.004 | 0.497 ± 0.028 | 0.502 | – |
| DP-GD LogReg ε = ∞ | 0.961 ± 0.009 | 0.021 | 0.503 ± 0.038 | 0.507 | – |
| DP-GD MLP-128 ε = ∞ | 0.961 ± 0.018 | 0.024 | 0.505 ± 0.022 | 0.507 | – |
| Input noise σ=0.25 | 0.956 ± 0.014 | 0.009 | 0.498 ± 0.025 | 0.511 | – |
| Input noise σ=0.5 | 0.904 ± 0.029 | -0.001 | 0.502 ± 0.027 | 0.508 | – |
| PCA k=2 | 0.943 ± 0.015 | 0.049 | 0.534 ± 0.015 | 0.511 | 0.544 |
| PCA k=5 | 0.961 ± 0.011 | 0.039 | 0.564 ± 0.018 | 0.510 | 0.355 |
| DCA k=1 | 0.961 ± 0.015 | 0.025 | 0.508 ± 0.027 | 0.513 | 0.721 |
| DCA k=5 | 0.941 ± 0.020 | 0.059 | 0.559 ± 0.034 | 0.524 | 0.732 |

WDBC is an easier and better-separated task (baseline accuracy 95.5%), so even the over-parameterised MLP generalises well and leaks much less (AUC 0.568). The deep random forest remains the leakiest model (AUC 0.604). The qualitative ordering is the same as on Pima: leakage tracks the generalisation gap, DP-GD and regularisation pull the attack to chance, and one DCA component keeps 96.3% accuracy while making reconstruction much harder (NRMSE 0.72 vs 0.54 for PCA with k = 2). **Membership risk is therefore a property of the model-data pair, not of the dataset alone.** A model that leaks badly on one small table can be reasonably safe on another.

## 4.9 Answers to the research questions

| RQ | Finding |
|---|---|
| RQ1 | Leakage tracks the generalisation gap (baseline AUC 0.77 at gap 0.28); regularisation cuts it to 0.49 with no significant accuracy change, and logistic regression is both the most accurate and least leaky model. |
| RQ2 | DP-GD at ε = 4 keeps 74.7% accuracy with attack AUC 0.51; the measured advantage stays far below the DP bound. The ablation shows that the short schedule DP requires causes much of the empirical gain; DP adds the guarantee. |
| RQ3 | LDP removes trust in the collector and suppresses MIA, but at n = 192 utility is limited (LDP-NB 66.8% at ε = 4); useful LDP-NB needs on the order of $10^4$ or more users. |
| RQ4 | Output restriction and k-anonymity only partially reduce MIA (label-only 0.67, k = 25 0.60); k-anonymity leaves homogeneity exposure that l-diversity removes. |
| RQ5 | One DCA component keeps 77.2% utility while age inference drops from 1.00 to 0.62 and reconstruction becomes infeasible; DUCA goes further for privacy. |
| RQ6 | Formal training-side protection (DP-GD) is the best overall defence for model release; compressive projections are the best defence for data release; syntactic and output-side measures are useful only as complements. |

## 4.10 Limitations and threats to validity

- **Internal validity.** The defended recipes were tuned lightly (e.g. DP learning rate, number of steps) and not by private hyperparameter search. Tuning on the data would itself consume privacy budget, so the reported ε excludes it.

- **Construct validity.** Attack AUC measures *average* leakage. Recent work (Carlini et al., 2022) argues for TPR at very low FPR, but with 192 members per run such low-FPR estimates are too noisy to report reliably. The attackers used are standard but not the strongest known (e.g. LiRA with many shadow models).
- **Conclusion validity.** With 10 seeds the smallest attainable two-sided Wilcoxon p-value is about 0.002. Small differences ($|\Delta$ AUC$| < 0.02$) should be read as 'no detectable difference', not as equivalence.
- **External validity.** Two small tabular datasets; Pima covers one population and one sex. Results may not transfer to large clinical datasets, images or language models.
- **DP accounting.** Full-batch DP-GD gives no subsampling amplification, so $\varepsilon$ is conservative, and add/remove adjacency is assumed. Public-bounds preprocessing is assumed to be truly public.
- **LDP scaling.** The user-scaling curves use bootstrap resampling of 576 records. They show the statistical trend, not real population diversity.
- **Not implemented.** Knowledge distillation (an IA-1 candidate) and DP hyperparameter selection were left out of scope.

## 4.11 Proposed improvements

- Add Poisson-subsampled DP-SGD with an RDP/PRV accountant to get privacy amplification and tighter $\varepsilon$.
- Evaluate LiRA and label-only augmentation attacks, and report TPR at FPR = 1% on a larger dataset (e.g. MIMIC-derived tabular data under a credentialed licence).
- Combine defences: DCA projection followed by DP-GD training, to protect both the released data and the model.
- Run a distillation (PATE-style) baseline to complete the IA-1 defence list.

## 5. Conclusion

This work turns the IA-1 membership inference attack into a common, quantitative yardstick and uses it to evaluate one defence family from each of the five course modules on a healthcare prediction task. Four conclusions follow. (i) Leakage is driven by memorisation: model choice and regularisation matter as much as any dedicated privacy technique. (ii) Central DP gives the best combination of utility, measured privacy and a provable guarantee for model release, provided ε is moderate (about 4). (iii) LDP and compressive privacy suit settings where the data holder or the data recipient is not trusted. LDP needs large user bases; DCA/DUCA can make released data nearly useless for attackers while keeping it useful for diagnosis. (iv) Syntactic anonymity and output restriction are weak against MIA and must be backed by l-diversity and formal methods. All code, data hashes and results are provided for reproduction, and an IEEE-style manuscript draft built on these results accompanies this report.

## 6. Ethics and Responsible Data Use

- Only public, de-identified datasets were used. No real healthcare service was queried, and no attempt was made to identify any individual.
- The Pima data come from the Akimel O'odham (Pima) community of the Gila River Indian Community. Its continued use as an ML benchmark has been criticised because of consent for secondary use and the risk of stigmatising a community. Here it is used only as a methodological benchmark, chosen for continuity with IA-1. No claims about the community are made, and WDBC (CC BY 4.0) is included so the conclusions do not depend on this dataset alone.
- The attack code is a measurement tool for evaluating defences. Releasing such tools follows standard practice in privacy research (e.g. TensorFlow Privacy's membership-inference tests).

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# Appendix A - Reproducibility

| Item | Value |
|---|---|
| Python / platform | 3.9.6 / macOS-26.6.2-arm64-arm-64bit |
| Key libraries | numpy 2.0.2, scikit-learn 1.6.1, scipy 1.13.1, pandas 2.3.3, matplotlib 3.9.4, reportlab 4.2.5 (pinned in requirements.lock.txt) |
| Seeds | 0-9 (split, model initialisation, DP noise and LDP noise all derive from the seed) |
| Shadow models | 8 per target, each trained on 192 pool records with 192 disjoint pool records as non-members |
| Raw data hash (Pima) | 6bfe5d0f379d17a0e0819b996407e3c09bf80febd4287f2ed212190dfff154af |
| Raw data hash (WDBC) | fed3eb72d0575ef6192293f5093c6e801b1476b577d0386bf4455504522172ed |
| Processed: pima_scaled | processed/pima_scaled.csv - 149e01c3b78bf2f67f4b2a7fd6d18b5c9b5e0196... |
| Processed: pima_discretised | processed/pima_discretised.csv - 4120d94497d3bac3a09526be74e71b6e8a91de21... |
| Processed: breast_cancer_scaled | processed/breast_cancer_scaled.csv - 07471ced7eb5817d1c23f8beaf0d2d2f83f1e570... |

Outputs: per-seed results `results/tables/mia_runs_*.csv`; aggregates `summary_*.csv`; paired tests `stats_*.csv`; standalone experiments `m2_*.csv`, `m3_*.csv`, `m4_anonymity.csv`, `m5_components.csv`; figures `results/figures/`; run log `results/run_log.txt`.

# Appendix B - Full Pima summary

| Configuration | API | Test acc | Test F1 | Shadow AUC | Shadow acc | Loss AUC | Loss adv |
|---|---|---|---|---|---|---|---|
| A: LogReg (L2) | full | 0.773 | 0.623 | 0.511 | 0.503 | 0.504 | 0.009 |
| B: MLP-32 | full | 0.762 | 0.636 | 0.571 | 0.546 | 0.548 | 0.101 |
| C: MLP-128x2 (baseline) | full | 0.719 | 0.583 | 0.765 | 0.688 | 0.631 | 0.380 |
| C: MLP-128x2 (baseline) | rounded | 0.719 | 0.583 | 0.707 | 0.681 | 0.681 | 0.362 |
| C: MLP-128x2 (baseline) | label | 0.719 | 0.583 | 0.666 | 0.641 | 0.641 | 0.281 |
| D: MLP-128x2 + L2 + early stop | full | 0.711 | 0.402 | 0.494 | 0.493 | 0.505 | 0.002 |
| E: Random forest (deep) | full | 0.759 | 0.623 | 0.798 | 0.724 | 0.766 | 0.419 |
| NumPy MLP-128, long training (non-private) | full | 0.700 | 0.566 | 0.741 | 0.673 | 0.615 | 0.327 |
| DP-GD LogReg ε=0.5 | full | 0.623 | 0.458 | 0.494 | 0.491 | 0.487 | -0.018 |
| DP-GD MLP-128 ε=0.5 | full | 0.557 | 0.341 | 0.519 | 0.510 | 0.508 | 0.014 |
| DP-GD LogReg ε=1 | full | 0.686 | 0.503 | 0.493 | 0.492 | 0.500 | 0.009 |
| DP-GD MLP-128 ε=1 | full | 0.594 | 0.365 | 0.498 | 0.493 | 0.505 | 0.003 |
| DP-GD LogReg ε=2 | full | 0.743 | 0.581 | 0.504 | 0.502 | 0.500 | 0.002 |
| DP-GD MLP-128 ε=2 | full | 0.687 | 0.481 | 0.500 | 0.495 | 0.510 | 0.000 |
| DP-GD LogReg ε=4 | full | 0.758 | 0.606 | 0.516 | 0.507 | 0.502 | 0.003 |
| DP-GD MLP-128 ε=4 | full | 0.747 | 0.566 | 0.510 | 0.499 | 0.511 | 0.025 |
| DP-GD LogReg ε=8 | full | 0.768 | 0.621 | 0.512 | 0.507 | 0.502 | 0.006 |
| DP-GD MLP-128 ε=8 | full | 0.758 | 0.595 | 0.518 | 0.509 | 0.509 | 0.020 |
| DP-GD LogReg ε=inf | full | 0.774 | 0.632 | 0.513 | 0.496 | 0.505 | 0.006 |
| DP-GD MLP-128 ε=inf | full | 0.767 | 0.630 | 0.517 | 0.512 | 0.515 | 0.024 |
| LDP-Piecewise LogReg ε=1 | full | 0.548 | 0.167 | 0.497 | 0.500 | 0.509 | 0.001 |
| LDP-Piecewise LogReg ε=4 | full | 0.619 | 0.035 | 0.508 | 0.510 | 0.509 | 0.002 |
| LDP-Piecewise LogReg ε=8 | full | 0.644 | 0.071 | 0.490 | 0.499 | 0.502 | 0.007 |
| LDP-Piecewise LogReg ε=16 | full | 0.657 | 0.111 | 0.497 | 0.497 | 0.500 | -0.006 |
| LDP Naive Bayes ε=1 | full | 0.578 | 0.370 | 0.497 | 0.493 | 0.505 | -0.005 |
| LDP Naive Bayes ε=2 | full | 0.609 | 0.424 | 0.513 | 0.510 | 0.496 | 0.011 |
| LDP Naive Bayes ε=4 | full | 0.668 | 0.410 | 0.514 | 0.508 | 0.501 | 0.008 |
| LDP Naive Bayes ε=8 | full | 0.691 | 0.480 | 0.506 | 0.498 | 0.506 | 0.007 |
| LDP Naive Bayes ε=inf | full | 0.752 | 0.648 | 0.525 | 0.511 | 0.529 | 0.030 |
| Mondrian k=5 | full | 0.702 | 0.555 | 0.675 | 0.634 | 0.617 | 0.262 |
| Mondrian k=10 | full | 0.709 | 0.577 | 0.647 | 0.608 | 0.596 | 0.222 |
| Mondrian k=25 | full | 0.694 | 0.563 | 0.598 | 0.576 | 0.576 | 0.149 |
| Mondrian k=10, l=2 | full | 0.714 | 0.580 | 0.640 | 0.609 | 0.595 | 0.210 |
| Input noise σ=0.25 | full | 0.701 | 0.535 | 0.565 | 0.543 | 0.548 | 0.086 |
| Input noise σ=0.5 | full | 0.689 | 0.475 | 0.521 | 0.511 | 0.521 | 0.037 |

| Configuration | API | Test acc | Test F1 | Shadow AUC | Shadow acc | Loss AUC | Loss adv |
|---|---|---|---|---|---|---|---|
| PCA k=2 | full | 0.699 | 0.482 | 0.538 | 0.529 | 0.528 | 0.056 |
| PCA k=4 | full | 0.705 | 0.566 | 0.695 | 0.646 | 0.597 | 0.267 |
| DCA k=1 | full | 0.745 | 0.607 | 0.537 | 0.534 | 0.535 | 0.050 |
| DCA k=2 | full | 0.722 | 0.592 | 0.651 | 0.610 | 0.577 | 0.220 |
| DCA k=4 | full | 0.691 | 0.557 | 0.783 | 0.701 | 0.626 | 0.398 |
| DUCA k=2 | full | 0.695 | 0.539 | 0.669 | 0.625 | 0.595 | 0.248 |